Lab - Calculate IPv4 Subnets

Objectives

Part 1: Determine IPv4 Address Subnetting

Part 2: Calculate IPv4 Address Subnetting

Background / Scenario

The ability to work with IPv4 subnets and determine network and host information based on a given IP address and subnet mask is critical to understanding how IPv4 networks operate. The first part is designed to reinforce how to compute network IP address information from a given IP address and subnet mask. When given an IP address and subnet mask, you will be able to determine other information about the subnet.

- 1 PC (Windows with Internet access)
- Optional: IPv4 address calculator

Instructions

Fill out the tables below with appropriate answers given the IPv4 address, original subnet mask, and new subnet mask.

Problem 1:

Given:	
Host IP Address:	192.168.200.139
Original Subnet Mask	255.255.255.0
New Subnet Mask:	255.255.255.224

Find:	
Number of Subnet Bits	3
Number of Subnets Created	8
Number of Host Bits per Subnet	**5**
Number of Hosts per Subnet	**30**
Network Address of this Subnet	**192.168.200.128**
IPv4 Address of First Host on this Subnet	**192.168.200.129**
IPv4 Address of Last Host on this Subnet	**192.168.200.158**
IPv4 Broadcast Address on this Subnet	**192.168.200.159**

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Problem 2:

Given:	
Host IP Address:	192.135.250.180
Original Subnet Mask	255.255.255.0
New Subnet Mask:	255.255.255.248

Find:	
Number of Subnet Bits	5
Number of Subnets Created	32
Number of Host Bits per Subnet	**3**
Number of Hosts per Subnet	**6**
Network Address of this Subnet	**192.135.250.176**
IPv4 Address of First Host on this Subnet	**192.135.250.177**
IPv4 Address of Last Host on this Subnet	**192.135.250.182**
IPv4 Broadcast Address on this Subnet	**192.135.250.183**

Problem 3:

Given:	
Host IP Address:	192.168.1.245
Original Subnet Mask	255.255.255.0
New Subnet Mask:	255.255.255.252

Find:	
Number of Subnet Bits	6
Number of Subnets Created	64
Number of Host Bits per Subnet	**2**
Number of Hosts per Subnet	**2**
Network Address of this Subnet	**192.168.1.224**
IPv4 Address of First Host on this Subnet	**192.168.1.245**
IPv4 Address of Last Host on this Subnet	**192.168.1.246**
IPv4 Broadcast Address on this Subnet	**192.168.1.247**

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Problem 4:

Given:	
Host IP Address:	128.107.0.55
Original Subnet Mask	255.255.0.0
New Subnet Mask:	255.255.255.0

Find:	
Number of Subnet Bits	8
Number of Subnets Created	256
Number of Host Bits per Subnet	**8**
Number of Hosts per Subnet	**254**
Network Address of this Subnet	**128.107.0.0**
IPv4 Address of First Host on this Subnet	**128.107.0.1**
IPv4 Address of Last Host on this Subnet	**128.107.0.254**
IPv4 Broadcast Address on this Subnet	**128.107.0.255**

Problem 5:

Given:	
Host IP Address:	172.22.32.12
Original Subnet Mask	255.255.0.0
New Subnet Mask:	255.255.224.0

Find:	
Number of Subnet Bits	3
Number of Subnets Created	8
Number of Host Bits per Subnet	**13**
Number of Hosts per Subnet	**8190**
Network Address of this Subnet	**172.22.32.0**
IPv4 Address of First Host on this Subnet	**172.22.32.1**
IPv4 Address of Last Host on this Subnet	**172.22.63.254**
IPv4 Broadcast Address on this Subnet	**172.22.63.255**

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Problem 6:

Given:	
Host IP Address:	10.101.99.228
Original Subnet Mask	255.0.0.0
New Subnet Mask:	255.255.128.0

Find:	
Number of Subnet Bits	9
Number of Subnets Created	512
Number of Host Bits per Subnet	**15**
Number of Hosts per Subnet	**32766**
Network Address of this Subnet	**10.101.0.0**
IPv4 Address of First Host on this Subnet	**10.101.0.1**
IPv4 Address of Last Host on this Subnet	**10.101.127.254**
IPv4 Broadcast Address on this Subnet	**10.101.127.255**

Reflection Question

Why is the subnet mask so important when analyzing an IPv4 address?

Subnetting splits on big network into smaller ones. In a normal IPv4 address there is no information about which part is part of the network, and which part is a part of the host. Subnetting draws the line for this where every 1 (in binary) is the network portion and every 0 (in binary) is the host portion.

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